

Justin Blackwood

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EDUCATION

University of Florida (GPA 3.51)

B.S. in Computer Science

Gainesville, Florida

Anticipated Dec 2027

EXPERIENCE

TDM Network Operations Specialist Intern

L3Harris Technologies

May 2026 – August 2026

Melbourne, Florida

- Performed remote fault isolation across national **FAA** network infrastructure spanning **24** airspaces, segmenting circuits with loopbacks between demarcation points, pulling alarm history from internal NMS tooling, and walking paths hop by hop to locate degraded spans.
- Triaged inbound tickets and received escalations across **TDM** transport multiplexers, DACS, Interplex units, T1/DS0 circuits, and carrier Ethernet, contributing to **100+** tickets and personally closing **25+**.
- Established fault ownership through remote testing and baseline comparison, attributing roughly **30%** of cases to customer-side equipment and redirecting them to FAA site checks rather than internal field dispatch.
- Pulled interface statistics from **Cisco** routers over **SSH**, coordinated findings with on-site technicians, and documented every resolution and shift handoff to preserve diagnostic continuity between operators.

Intro to Programming Teaching Assistant

University of Florida

August 2025 – May 2026

Gainesville, Florida

- Wrote half of the **Python** autograding scripts for the course on an inherited grading platform, covering the full introductory curriculum from standalone functions through class-based assignments and processing several hundred submissions per assignment.
- Leading two weekly Programming I lab sessions of **60** students plus office hours for **20+** first-year CS students, covering foundational concepts, **Python** labs, and hands-on debugging.
- Contributing observations from lab and office-hour sessions in weekly faculty planning meetings that informed course restructuring.

PROJECTS

M.I.R.A. a Local AI Assistant | *Python, Ollama, Tkinter*

June 2026 – Present

- Building a local-first assistant as a persistent background daemon, split into separate chat, model-dispatch, and tool-registry layers so conversation handling, prompt/history management, and side effects stay independently testable.
- Architected a two-model router/responder split, then diagnosed a context-staleness defect caused by classifying each message in isolation and redesigned around native tool calling so the responder evaluates full conversation history before dispatching a tool.
- Implemented a tool registry with parsed-argument dispatch and persistent tool-result turns, plus per-turn injection of live state into a replaced system slot so reads never depend on classifier accuracy.

Draft League Bot | *Python, JSON, GitHub*

July 2025 – February 2026

- Designed and deployed a custom bot automating drafting, point tracking, and rule enforcement, reducing admin work by **80%** and now in active use by **25** league members.
- Coordinated a team of **10** testers through structured testing sessions, managing bug tracking and resolution to identify and resolve **7** critical defects before release.
- Collaborated with a contributing developer via GitHub to review, integrate, and test new features while maintaining consistency with existing systems and league rules.

TECHNICAL SKILLS

Languages: Python, C++, Typescript

Developer Tools: Git/GitHub, VS Code, Linux, PuTTY/SSH, Cloudflare

Libraries: SFML, discord.py, Pygame, socket

Technologies: JSON, Node.js, SQLite, React.ts, Ollama

Networking: TDM transport, T1/DS0, carrier Ethernet, DACS, Cisco IOS

Relevant Courses: Data Structures & Algorithms, Computer Organization, Computer Networking, Operating Systems